WELCOME TO SOLAIRE INTELLIGENCE

Let’s protect your pool pumps.

SI Pool keeps a constant eye on all three pool pumps through a small Shelly power meter on each pump. If a suction line collapses or a pump starts running dry, the power signature changes long before the motor is damaged — and we cut the pump automatically.

HOW IT FITS TOGETHER

IN THE SI POOL BOX

Brain

- **SI Pool** — Raspberry Pi (pre-configured)
- SSD storage (pre-loaded with SI software)
- USB-C power supply
- Ethernet patch cable

SHELLY SWITCH BOXES

Muscle (x3)

- **3 x Shelly Plus 1PM** (Gen 4)
- One switch per pump
- Rated 16 A / 240 V~ single-phase
- Shelly screws / strain relief in box

WHAT YOU’LL NEED

Before you start

- A **competent person** to isolate and wire the pumps
- Voltage tester (non-contact + two-pole)
- Your site Wi-Fi name and password (2.4 GHz)
- Phone with the **Shelly** app installed
- Network cable to the router or switch

AT A GLANCE

The 6 steps ahead

1	Unpack & identify the kit
2	Wire a Shelly 1PM into each pump
3	Add the three Shellys to Wi-Fi and name them Pump 1, 2, 3
4	Plug in and boot the SI Pool Pi
5	Open the system in your browser
6	We take it from here — dashboard + dry-run guardrails

1 Unpack & identify the kit

Lay the kit out on a clean, dry surface in the pool plant room or workshop. Check each box against the lists on page 1 before you start wiring.

- **SI Pool** — the small box with a Raspberry Pi and USB-C power supply inside.
- **3 x Shelly Plus 1PM (Gen 4)** — one goes inline with each pump.
- Check the **label on the Pi itself** — it carries the **unique Home Assistant username and password** for this SI Pool. You'll need them in Step 5.

> **Three pumps, three switches — no sharing**
>
> Each pump must have its own Shelly 1PM. We need to meter each pump **independently** so we can spot the one that's starving and shut just that one down, instead of killing the whole pool plant.

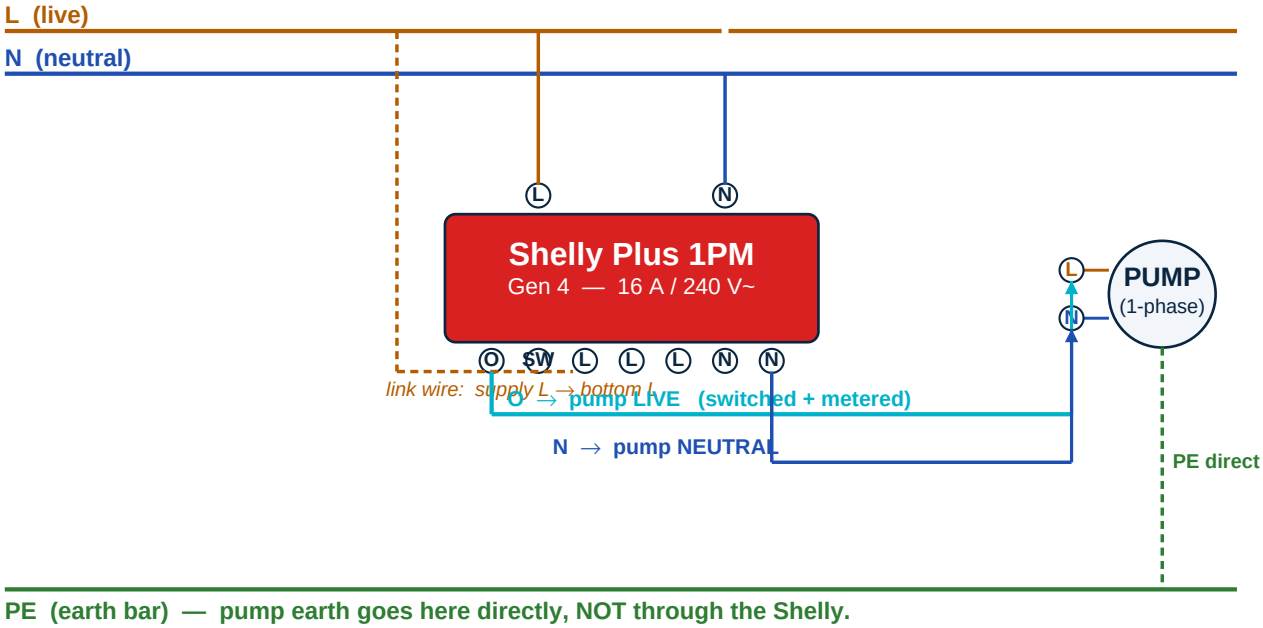
2 Wire a Shelly 1PM into each pump

> **Isolate before you open anything**
>
> Before touching any wiring: switch the pump circuits **OFF at the distribution board**, lock or tape the breaker in the OFF position, and confirm with a **voltage tester** that L and N are dead at the pump terminals. If you are not comfortable working on 230 V mains, stop here and call a qualified electrician.

The Shelly sits **between the DB and the pump**, in line with the pump's live. It switches the pump on and off, and measures current and power so SI Pool can see exactly what each pump is doing. Repeat the same wiring for all three pumps — do one at a time, mark it, then move on.

Shelly Plus 1PM — wiring, one per pump (230 V~ single-phase)

From the pump's MCB / isolator at the DB (repeat for Pump 1, Pump 2 and Pump 3)



Terminal	Role	Wire to
L (top)	Supply live in	From the pump circuit breaker (MCB) at the DB.
N (top)	Supply neutral in	From the neutral bar at the DB.
O	Switched + metered live OUT	Goes to the pump's **L** terminal. This is the wire the Shelly switches and monitors.
N (bottom)	Neutral OUT to pump	Goes to the pump's **N** terminal.
L (bottom)	Internal L jumper	Connect one bottom L to the supply L feeding the top so the Shelly has a reference for the switched output. See the diagram.
SW	Manual override (optional)	Leave empty. If the site wants a local on/off switch later, we'll wire it here.
PE / earth	NOT through the Shelly	Pump earth goes **directly** to the earth bar in the DB. Never route earth through the Shelly.

■ Do one pump at a time

Isolate, wire, close up, label the Shelly (e.g. masking tape marked **P1**, **P2**, **P3**), then move to the next pump. It makes Step 3 far easier — and makes sure that the Shelly you're naming "Pump 1" really is pump 1.

3 Add the Shellys to Wi-Fi and name them

Once the three Shellys are wired and the breakers are back on, use the free **Shelly** app on your phone to add each one to the site Wi-Fi and give it the right name.

- Install the **Shelly** app from the App Store / Play Store and create a free account (or sign in).
- Tap **+ Add device**. The app will search for any Shelly nearby that's in pairing mode.
- Energise **only one Shelly at a time** (switch on just that pump's breaker) so there's no confusion about which one you're pairing.
- When the app finds it, select your site's **2.4 GHz Wi-Fi** and enter the password.
- When prompted for a name, type **Pump 1** (exactly, with the space — we match on this name).
- Repeat for the other two, naming them **Pump 2** and **Pump 3**. Energise them one at a time.

Band	**2.4 GHz only** — Shellys don't speak 5 GHz
Network	The **same LAN** as the SI Pool Pi. Don't put the Shellys on a guest network.
Names (exact)	`Pump 1` `Pump 2` `Pump 3`

■ Naming matters

SI Pool looks for the names **Pump 1**, **Pump 2** and **Pump 3** on the network. If you call one "Pump1" (no space) or "Main pump", we won't find it and the dry-run guardrails won't arm on that pump. If you're not sure which Shelly is which, rename it to match the tape label you put on the enclosure in Step 2.

■ Quick sanity check before Step 4

In the Shelly app, tap each Pump in turn and toggle the output OFF, then ON. You should hear each pump stop and start. If a different pump reacts than the one you tapped, two labels are swapped — fix the names now, not later.

4 Plug in and boot the SI Pool Pi

The SI Pool is a Raspberry Pi already flashed with the Solaire Intelligence image and connected to its SSD. You just need to give it power and put it on the same network as the Shellys.

- Connect an Ethernet patch cable from the Pi to your **router or network switch**.
- Plug the USB-C power supply into the Pi and into a wall outlet.
- Wait **2–3 minutes** for the first boot to complete.

> ■ **Same LAN as the Shellys**
>
> SI Pool finds the three Shelly switches by listening on the local network. The Pi **must** be on the same LAN / subnet as the Shellys. Wired Ethernet is required — Wi-Fi is not supported for the Pi.

5 Open the system in your browser

Once the Pi has finished booting, you can reach the dashboard from any computer, phone or tablet on the same network.

Open a browser and go to	`http://homeassistant.local:8123`
Log in with	the **username and password printed on the label on your Pi**. Each SI Pool ships with its own unique credentials — they’re not shared between units.

> ■ **Can’t reach homeassistant.local?**
>
> Some older Windows PCs and certain routers don’t resolve **.local** names. As a fallback, look in your router’s DHCP client list for a device called **homeassistant** or **raspberrypi** and use its IP address instead, e.g. **http://192.168.1.42:8123**.

6 We take it from here

With all three Shellys on Wi-Fi and the SI Pool Pi on the LAN, **no further action is required from you**. The SI team connects to your system remotely and finishes the job.

What happens	Detail
Pump discovery	SI Pool will detect Pump 1, Pump 2 and Pump 3 on the LAN and start reading live current, voltage and power from each.
Baseline & guardrails	We watch each pump for a day or so to learn its normal running power. Then we arm **dry-run protection**: if a pump's power drops below its running band (suction flat / prime lost) or spikes (strainer blocked), SI Pool **cuts that pump** through its Shelly and alerts you.
Dashboard & handover	You'll see a live tile per pump, a per-day runtime / energy total, and any alarms. We send you a short welcome email with the dashboard link and a tour when it's live.

TROUBLESHOOTING

If something doesn't look right

Symptom	Try this
A Shelly doesn't appear in the Shelly app	Confirm that pump's MCB is on. If it's never been paired, hold the button on the Shelly for 5 s to force pairing mode, and try again with only that pump energised.
SI Pool can't see a Shelly that's on Wi-Fi	Make sure the Shelly is on the **same network/VLAN** as the Pi (not on a guest Wi-Fi) and the name is exactly **Pump 1** / **Pump 2** / **Pump 3**.
A pump reads 0 W even when running	Power is reaching the pump but not via **O**. The pump live is on a bottom **L** terminal by mistake. Re-check Step 2 wiring on that Shelly.
Pump trips off on its own	That's the dry-run guardrail doing its job. Check strainer baskets, skimmer weirs and the suction line. The dashboard alarm names the pump and direction.

Welcome to the SI family. You're now running smarter, safer pool plant. If anything looks off in the first 24 hours, drop us a line — we'll check it remotely.

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